

## SDE Intern Coding Exam

Coding

Q1 of 3

## Q1 Optimal Array Partition

10 marks

2s limit

256MB

You are given an array  $A$  of  $N$  positive integers. You need to partition this array into exactly  $K$  non-empty contiguous subarrays such that the **cost** of the partition is minimized.

The **cost** of a partition is defined as the sum of the **costs** of each subarray, where the cost of a subarray is calculated as:

- $\text{cost}(\text{subarray}) = (\text{max\_element} - \text{min\_element}) * \text{length\_of\_subarray}$

In other words, for each subarray, you multiply the range (difference between maximum and minimum elements) by the number of elements in that subarray.

## Task

Find the minimum possible cost to partition the array into exactly  $K$  contiguous subarrays.

## Input Format

- First line contains two space-separated integers  $N$  and  $K$ .
- Second line contains  $N$  space-separated integers

&lt;&gt; Solution

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     // Your code here
6     return 0;
7 }
```

Output





## SDE Intern Coding Exam

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Q1 of 3

## Q1 Optimal Array Partition

10 marks

2s limit

256MB

- First line contains two space-separated integers  $N$  and  $K$ .
- Second line contains  $N$  space-separated integers representing array  $A$ .

## Output Format

- Print a single integer - the minimum cost to partition the array into exactly  $K$  subarrays.

## Examples

## Example 1

Input:

```
5 2
1 3 2 5 4
```

Output:

```
8
```

&lt;&gt; Solution

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      // Your code here
6      return 0;
7  }
```

Output





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Explanation: Optimal partition is [1, 3, 2] and [5, 4] :

- Cost of [1, 3, 2] =  $(3 - 1) * 3 = 6$
- Cost of [5, 4] =  $(5 - 4) * 2 = 2$
- Total cost =  $6 + 2 = 8$

## Example 2

Input:

4 4

1 2 3 4

Output:

0

Explanation: Each element in its own subarray: cost of single element =  $(x - x) * 1 = 0$ .

Note

&lt;&gt; Solution

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      // Your code here
6      return 0;
7  }
```

Output



**Q1** Optimal Array Partition

10 marks

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## Example 1

Input:

5 2  
1 3 2 5 4

Output:

8

## Example 2

Input:

4 4  
1 2 3 4

Output:

0

&lt;&gt; Solution

```
5  int main
--
20     for(
21
22
23
24
25
26
27
28
29
30
31
32
33
```

Output 1/2 s

✓ Sample Test

✗ Sample Test

Error:

Wrong Answer

Expected:

0



## SDE Intern Coding Exam

Coding

Q2 of 3

## Q2 Magical Bridges

10 marks

3s limit

256MB

## Problem Statement

In the kingdom of Algorithmia, there are  $N$  cities numbered from 1 to  $N$ , connected by  $M$  bidirectional roads. Each road has a certain travel time associated with it.

The kingdom also has  $K$  magical bridges. Each magical bridge connects two cities and has zero travel time, but each magical bridge can only be used **once** during your entire journey. You cannot use more than  $K$  magical bridges in total.

You are a traveler starting from city 1 and want to reach city  $N$ . Find the minimum time required to travel from city 1 to city  $N$  using at most  $K$  magical bridges.

## Input Format

- First line contains three space-separated integers  $N$ ,  $M$ , and  $K$ .
- Next  $M$  lines each contain three integers  $u$ ,  $v$ ,  $w$  representing a road between cities  $u$  and  $v$  with travel time  $w$ .

&lt;&gt; Solution

```
1 #i
2 us
3
4 in
5
6
7 }
```

Output



## Q2 Magical Bridges

10 marks

3s limit

256MB

## Output Format

- Print a single integer - the minimum time to reach city N from city 1.
- If it is impossible to reach city N, print -1.

## Examples

## Example 1

Input:

```
4 4 1
1 2 10
2 4 10
1 3 5
3 4 20
1 4
```

Output:

&lt;&gt; Solution

```
1 #include <iost
2 using namespac
3
4 int main() {
5     // Your co
6     return 0;
7 }
```

Output



## Q2 Magical Bridges

10 marks

3s limit

256MB

Output:

0

Explanation: Use the magical bridge from city 1 directly to city 4, taking 0 time.

## Example 2

Input:

```
4 4 1
1 2 10
2 4 10
1 3 5
3 4 20
2 3
```

Output:

15

&lt;&gt; Solution

```
1 #include
2 using n
3
4 int mai
5 //
6 ret
7 }
```

Output



## Q2 Magical Bridges

10 marks

3s limit

256MB

Output:

15

Explanation: Optimal path: 1 → 3 (time 5) → magical bridge to 2 (time 0) → 4 (time 10). Total = 15.

## Example 3

Input:

```
3 1 0
1 2 10
```

Output:

-1

Explanation: No path exists from city 1 to city 3 without magical bridges.

## Note

&lt;&gt; Solution

```
1 #incl
2 using
3
4 int m
5 /
6 r
7 }
```

Output



## Q2 Magical Bridges

10 marks

3s limit

256MB

-1

**Explanation:** No path exists from city 1 to city 3 without magical bridges.

**Note**

- Magical bridges are bidirectional.
- You can choose to not use all  $K$  magical bridges.
- Regular roads can be used multiple times.

**Constraints**

$2 \leq N \leq 10^4$   $1 \leq M \leq 5 * 10^4$   $0 \leq K \leq 10$   $1 \leq u, v, a, b \leq N$   $1 \leq w \leq 10^6$  Time Limit: 3 seconds  
Memory Limit: 256 MB

**Sample Test Cases**

Example 1

&lt;&gt; Solution

```
1 #inclu
2 using
3
4 int ma
5 //
6 re
7 }
```

Output



## SDE Intern Coding Exam

Coding

Q2 of 3

## Q2 Magical Bridges

10 marks

3s limit

256MB

Sample Test Cases

## Example 1

Input:

```
4 4 1
1 2 10
2 4 10
1 3 5
3 4 20
1 4
```

Output:

0

## Example 2

Input:

```
4 4 1
1 2 10
2 4 10
- - -
```

&lt;&gt; Solution

```
1 #include
2 using n
3
4 int mai
5 //
6 ret
7 }
```

⚙️ Output



## Q2 Magical Bridges

10 marks

3s limit

256MB

### Example 2

Input:

```
4 4 1
1 2 10
2 4 10
1 3 5
3 4 20
2 3
```

Output:

15

### Example 3

Input:

<> Solution

```
11 int main()
12
13 ll mn_
14 for(in
15 in
16 if
17
18 }
19 }
20 for(i
21 i
22 i
23 i
24
25
```

⚙ Output 2/3 s

✓ Sample Test

✗ Sample Test

Error:



## SDE Intern Coding Exam Coding Q3 of 3

28:58

Submit &amp; End Exam

## Q3 Prime Subsequence Sum

10 marks 3s limit 256MB

You are given an array  $A$  of  $N$  positive integers. You need to find the number of non-empty subsequences of the array such that the sum of the subsequence is a prime number.

Since the answer can be very large, print it modulo  $10^9 + 7$ .

## Definitions

- A **subsequence** is a sequence derived from the array by deleting zero or more elements without changing the order of remaining elements.
- A **prime number** is a natural number greater than 1 that has no positive divisors other than 1 and itself.

## Input Format

- First line contains a single integer  $N$ .
- Second line contains  $N$  space-separated integers representing array  $A$ .

## Output Format

&lt;&gt; Solution

C++

Run

Submit Code

```
1 #include <bits/stdc++.h>
2 using namespace std;
3
4 const int MOD = 1000000007;
5
6 int main() {
7     ios::sync_with_stdio(false);
8     cin.tie(NULL);
9
10    int N;
11    cin >> N;
12
13    vector<int> A(N);
14    int totalSum = 0;
```

Output 10/10 test cases passed



Accepted!

Your solution passed all test cases

Previous Question

Questions

(1 suggested)

1 2 3

Answered

Visited

Not Visited